

PMBOK / PMI | PMB-15

Cost Estimates and Basis of Estimates

Cost estimates for each work package, with the reasoning that supports them

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Cost Estimates and Basis of Estimates
Document reference	PMB-15-HEL-T2-v1.0
Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	Project manager with work package owners
Approver	Sponsor accepts the estimate as the basis for the budget
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
PMB-14	Cost Management Plan	Governs estimating
PMB-16	Cost Baseline and Funding Requirements	Consumes these estimates
PMB-22	Risk Register and Risk Report	Source of contingency

How to use this template

Aspect	Guidance
Purpose	Records the estimated cost of each activity or work package together with the basis of estimate — the method, source data, assumptions, constraints, confidence range and the risks reflected in the estimate.
Framework	PMBOK / PMI — Cost Management — Estimate Costs
Created when	During planning; refined at each phase gate
Reviewed / updated	When the estimate class changes or significant new information arrives
Owner	Project manager with work package owners
Approver	Sponsor accepts the estimate as the basis for the budget
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	The basis of estimate matters more than the number. An estimate without a basis cannot be challenged, defended or improved.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- Every estimate has a stated method and source
- Ranges are given, not single-point figures
- The estimate class is stated and matches the level of definition available
- Assumptions and exclusions are explicit
- Contingency is derived from risk analysis, not applied as a flat percentage

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1. Estimate summary

Field	Detail
Estimate version	v1.0
Estimate class	Verified at hall A acceptance on 26 Feb 2027
Date prepared	18 Sep 2026
Prepared by	D. Okonkwo, Programme Manager
Accuracy range	-5% to +10% for the current tranche
Base cost total	AED 118,000,000 for Tranche 2; tolerance of +5% before escalation
Contingency	Carried forward from the Tranche 1 lessons register
Management reserve	AED 5,200,000, held outside the cost baseline and released by the SRO only
Total	AED 986,000

2. Cost estimates by work package

WBS code	Work package	Category	Quantity	Rate	Low	Expected	High	Method
1.0	Programme management	Internal labour	4,200 h	AED 1,050	AED 3,900,000	AED 4,410,000	AED 5,100,000	Bottom-up from role rates
2.0	Facilities and power	Facilities	3,100 h	AED 1,180	AED 21,700,000	AED 24,300,000	AED 29,800,000	Contractor quotation plus 12% risk
2.1	Utility uplift to 6 MVA	Facilities	640 h	AED 1,180	AED 2,900,000	AED 3,400,000	AED 4,200,000	Utility schedule of rates
2.2	Liquid cooling distribution	Facilities	980 h	AED 1,180	AED 8,600,000	AED 9,700,000	AED 12,100,000	M&E quotation
2.3	Structural strengthening	Facilities	520 h	AED 1,180	AED 2,100,000	AED 2,400,000	AED 3,100,000	Structural engineer estimate
3.0	Compute and storage	Hardware	2,600 h	AED 1,120	AED 100,800,000	AED 109,000,000	AED 119,500,000	Vendor quotation, firm to Dec 2026
3.1	Accelerator install and burn-in	Hardware	1,480 h	AED 1,120	AED 84,600,000	AED 91,200,000	AED 99,400,000	Vendor quotation for 2,048 units
3.2	Parallel filesystem build	Hardware	720 h	AED 1,120	AED 16,200,000	AED 17,800,000	AED 20,100,000	Vendor quotation, 12

WBS code	Work package	Category	Quantity	Rate	Low	Expected	High	Method
					0,000	000	0,000	PB usable
4.0	Network fabric	Network	1,900 h	AED 1,140	AED 12,400,000	AED 13,600,000	AED 15,900,000	Vendor quotation plus cabling rates
4.1	Fabric build and cabling	Network	1,100 h	AED 1,140	AED 10,900,000	AED 11,800,000	AED 13,600,000	Bill of materials plus install rates

3. Basis of estimate

WBS code	Method used	Source data	Key assumptions	Confidence
Consolidation is the primary source of the benefit case; utilisation across siloed estates averages 31%	Live walkthrough with a recorded session	Programme SharePoint — Helios / Tranche 2 / Governance	2,048 units including 5-year vendor support	Low
Air cooling would cap density at 12 kW per rack and require 2.4 times the floor area	Written pack circulated 48 hours in advance	Configuration management database	12 PB usable, 1.2 TB/s sustained	Medium
Two independent schedulers fragment the GPU pool and reduce effective capacity	Automated report from the observability platform	Assurance evidence pack, Tranche 2 gate	400G leaf-spine, optics and structured cabling	High
Air	Workshop with	ALM tool — Helios	CDUs, rear-door heat exchangers,	Low

WBS code	Method used	Source data	Key assumptions	Confidence
cooling would cap density at 12 kW per rack and require 2.4 times the floor area	the affected workstream leads	ART board	floor strengthening, 6 MVA uplift	
Consolidation is the primary source of the benefit case; utilisation across siloed estates averages 31%	Dashboard published to the programme site	Solution intent wiki, DXB-1 platform space	Orchestration, observability, image supply chain, MLOps tooling	Medium
Supply chain provenance for AI tooling is weak; admission control is the only enforceable point	Live walkthrough with a recorded session	Programme SharePoint — Helios / Tranche 2 / Governance	Design, build and commissioning across three tranches	High
Grid capacity at DXB-1 supports Tranche 2 in full, so a second site is not yet justified	Written pack circulated 48 hours in advance	Configuration management database	34 FTE-years across eight workstreams	Low
Physical lead times do not tolerate emerging requirements, so	Automated report from the observability platform	Assurance evidence pack, Tranche 2 gate	Penetration testing, control assurance, supply chain verification	Medium

WBS code	Method used	Source data	Key assumptions	Confidence
this element is frozen before order				

4. Estimate exclusions

What the estimate does not cover. This section prevents the most damaging kind of budget surprise.

Excluded	Rationale	Who bears the cost
Confirmed at the Tranche 1 end-of-tranche review, 18 Sep 2026	2,048 units including 5-year vendor support	AED 1,120,000
Enforced automatically in the delivery pipeline, with evidence retained	12 PB usable, 1.2 TB/s sustained	AED 17,800,000
Reviewed at each PI boundary by the Release Train Engineer	400G leaf-spine, optics and structured cabling	AED 240,000
Escalated to the Portfolio Board where it exceeds programme tolerance	CDUs, rear-door heat exchangers, floor strengthening, 6 MVA uplift	AED 11,200,000
Maintained by the P3O as part of the assurance evidence library	Orchestration, observability, image supply chain, MLOps tooling	AED 96,000
Aligned to the strategic theme of sovereign AI capability	Design, build and commissioning across three tranches	AED 4,200,000

5. Contingency derivation

Derived from the risk register. Show the working.

Risk ref	Risk	Probability	Cost impact	Expected value
Weak tenant isolation exposes model weights across business units (RSK-003)	Scarce platform engineering skills extend contractor dependence (RSK-004)	Constrained by the 6.2 MVA site limit at DXB-1 until 2028	AED 11,200,000	AED 4,200,000
Hall B cooling capacity limits achievable rack	Vendor allocation slippage defers the Tranche 2 benefit by a quarter (RSK-001)	Aligned to the strategic theme of sovereign AI capability	AED 240,000	AED 96,000

Risk ref	Risk	Probability	Cost impact	Expected value
density (RSK-002)				
Vendor allocation slippage defers the Tranche 2 benefit by a quarter (RSK-001)	Hall B cooling capacity limits achievable rack density (RSK-002)	Agreed with the integration partner at design freeze, 15 Oct 2026	AED 17,800,000	AED 11,200,000
Scarce platform engineering skills extend contract or dependence (RSK-004)	Weak tenant isolation exposes model weights across business units (RSK-003)	Owned within WS4 Platform and orchestration	AED 1,120,000	AED 240,000

6. Rate card used

Resource type	Rate	Unit	Source	Valid until
ALM tool — Helios ART board	AED 240,000	Confirmed with the FM provider under the existing facilities contract	Configuration management database	15 Dec 2026
Solution intent wiki, DXB-1 platform space	AED 17,800,000	Covers the 2,048-GPU estate and the parallel filesystem tier	Assurance evidence pack, Tranche 2 gate	30 Nov 2026
Programme SharePoint — Helios / Tranche 2 / Governance	AED 1,120,000	Reviewed at each PI boundary by the Release Train Engineer	ALM tool — Helios ART board	15 Oct 2026

Resource type	Rate	Unit	Source	Valid until
Configuration management database	AED 91,200,000	Assured independently by Cordant Assurance	Solution intent wiki, DXB-1 platform space	18 Sep 2026

7. Estimate review and challenge

Date	Reviewer	Challenge raised	Response	Change made
30 Nov 2026	15 Dec 2026	31 Jan 2027	Run two live coolant-leak drills before hall A handover	Bridge the hall B row 3 service void rather than relocate racks
15 Dec 2026	31 Jan 2027	26 Feb 2027	Enforce mutual TLS on the cluster API and retest by 22 Sep 2026	Dual-source 20% of Tranche 2 accelerator volume
31 Jan 2027	26 Feb 2027	31 Mar 2027	Enforce mutual TLS on the cluster API and retest by 22 Sep 2026	Add coolant leak detection integration to the NOC
26 Feb 2027	31 Mar 2027	23 Apr 2027	Run two live coolant-leak drills before hall A handover	Reduce Tranche 2 GPU count from 2,048 to 1,792
31 Mar 2027	23 Apr 2027	28 May 2027	Broker all vendor access through the monitored in-country jump host	Add a second CDU to hall A to reach the 60 kW design point